Graph-Guided Retrieval for Textbook-based RAG Systems

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Mcgill University
Shiyuan Qiao 260967649
Jeremy Piperni 260990011
Cloris Liu 261010926

Abstract

Retrieval-Augmented Generation (RAG) systems are becoming increasingly important in helping large language models generate precise and well-supported responses. However, educational materials present a unique challenge for standard embedding-based retrieval due to their long-range dependencies and hierarchical structure. This project investigates whether textbook metadata can be used to build a lightweight knowledge graph that improves retrieval accuracy for both detailed questions and chapter summaries. Using the OpenStax Psychology 2e textbook (Fartale et al., 2024) as a case study, we extract hierarchical metadata from its table of contents and match each page to its corresponding chapter and section. This creates a directed graph that captures parent–child and sibling relationships between sections. Based on this graph, we propose a graph-guided retrieval framework that starts with vector search and then expands candidate nodes along structural neighbours before re-ranking them. Our tests with the MiniLM model show significant improvements in recall, hit rate, and gold-page coverage across 32 QA queries and 12 summarization prompts, with low latency. Enhancements are especially notable for summary questions, which require synthesizing information from multiple sections. These results show that hierarchical metadata provides an effective, efficient signal for improving retrieval in RAG systems, especially in education.

1 Introduction

Retrieval-Augmented Generation (RAG) has become a key method for providing factual grounding in large language models. RAG systems retrieve relevant passages from a corpus and input them into a generator. The success of RAG heavily depends on the quality of retrieval. If retrieval provides irrelevant, incomplete, or noisy results, generated answers become unfaithful or inaccurate. This problem is well documented in recent evaluations, where retrieval errors are identified as a main failure mode in live RAG implementations. This challenge is particularly relevant in educational contexts, where textbooks have long-range dependencies and hierarchical structures. Precise retrieval from textbooks is essential for creating effective explanations, and the understanding developed here could also apply to other structured texts such as technical manuals or legal documents.

An increasing number of studies show that metadata such as chapter titles, timestamps, section labels, or domain-specific structural cues can improve retrieval by providing additional signals beyond raw embeddings. However, most RAG systems currently treat metadata as simple annotations rather than structured relationships. Textbooks, especially, have a clear hierarchical structure: chapters contain sections, and sections often share conceptual similarities. Modeling this hierarchy could improve the retriever's ability to handle ambiguous or broad-ranging queries more effectively. Our main innovation lies in using this hierarchy not only for filtering but also as an active graph to expand and re-rank queries dynamically during runtime.

This project investigates the following research question: **Can modelling textbook metadata as a knowledge graph improve retrieval accuracy and contextual faithfulness in generated answers?** We expect this investigation to reveal how much structural signals contribute beyond semantic similarity and under which conditions graph expansion provides the greatest benefit.

To answer this, we develop a graph-guided retrieval module that expands candidate results using the graph neighbourhood before re-ranking them. We evaluate performance on a collection of 32 educational questions drawn from the Psychology 2e textbook (Fartale et al., 2024). We extract hierarchical metadata from the PDF textbook by parsing its table of contents and assigning each page to its corresponding chapter and section labels. Using this structure, we build a knowledge graph that captures parent–child relationships between chapters and sections, as well as sibling links between sections within the same chapter. Leveraging this graph, we develop a graph-guided retrieval mechanism that begins with embedding-based retrieval, then expands the candidate set through structurally related nodes before re-ranking them. Our empirical evaluation shows that this metadata-aware retrieval approach significantly improves recall, hit-rate, and contextual grounding, while adding only minimal latency overhead.

2 Related Work

Recent surveys on RAG systems highlight three main failure areas: inconsistent retrieval quality, limited interpretability, and sensitivity to noisy documents (Oche et al., 2025). Many attempts to improve faithfulness focus on prompt engineering, model fine-tuning, or dataset filtering, but fewer explore structural metadata.

Metadata-driven retrieval has demonstrated promise in early studies. (Bruni et al., 2025) show that structured metadata, such as chapter labels, section identifiers, and semantic groupings, can improve retrieval accuracy, especially in QA tasks. Their dataset, AMAQA, includes metadata for re-ranking but does not model relationships between sections. Meanwhile, production-focused RAG analyses highlight that reliability often depends on controlling context selection, especially within large knowledge bases (Bouchard and Peters, 2024). These studies encourage developing retrieval components that use explicit structure in addition to simple vector similarity.

Our work expands on these insights by constructing a lightweight hierarchical knowledge graph directly from PDF metadata and incorporating it into a retrieval pipeline. Unlike prior metadata-based filtering, our method uses graph expansion to propagate contextual signals across related sections. This allows us to leverage the natural parent–child and sibling relationships in textbooks, which prior research has mostly regarded as simple annotations.

3 Data and Environment

3.1 Data

We use the OpenStax Psychology 2e textbook (PDF) (Fartale et al., 2024) as the main retrieval corpus. The textbook is processed page-by-page using `pdfplumber` and `PyPDF2`, resulting in 751 normalized page segments. Each page is aligned with hierarchical metadata extracted from the PDF outline, including chapter titles, section headings, and section numbering patterns. This alignment allows every page to be annotated with a `chapter` and `section` field, which later becomes the basis of the metadata graph used for retrieval expansion. In addition to the corpus, we use a JSON dataset containing 32 fine-grained question–answer pairs and 12 chapter-level summarization prompts. Each query is annotated with gold-standard page spans that we use as ground truth for retrieval evaluation.

3.2 Evaluation Metrics

We evaluate retrieval quality using four standard metrics commonly used in RAG and information retrieval. Recall@k measures the proportion of gold-standard pages or chunks retrieved within the top–k results. Hit rate@k shows whether at least one relevant item appears in the top–k for a given query, providing a query-level measure of coverage. Mean Reciprocal Rank (MRR) indicates how early the first relevant item appears in the ranked list, rewarding systems that rank relevant evidence near the top.

Finally, we report the gold-ratio@k, the proportion of all gold items successfully retrieved, which provides a more detailed view of completeness in multi-label settings such as chapter-level summarization. All metrics are computed for both fine-grained QA retrieval at the chunk level ($k = 5$) and summary-style retrieval at the page level ($k = 20$).

3.3 Experimental Environment

All experiments are implemented in Python 3.9. We use the `sentence-transformers` library with the `all-MiniLM-L6-v2` model for generating embeddings (384-dimensional vectors). Retrieval is performed using `ChromaDB` as the vector store. Graph construction and manipulation use `NetworkX` for representing the hierarchical metadata graph.

Experiments were run on a machine with an NVIDIA RTX 3060 Ti GPU (8GB VRAM) and 32GB RAM. The total indexing time for the 751-page corpus was approximately 45 seconds, and average query retrieval latency was under 15 ms. All code and processed data are available in our supplementary repository to ensure reproducibility.

4 Method

Our research hypothesis is that modelling textbook metadata as a hierarchical knowledge graph improves retrieval accuracy and contextual understanding more effectively than using dense embeddings alone can provide. In particular, we expect that the chapter–section structure supplies relational cues that help identify conceptually coherent content, particularly for queries that involve information distributed across multiple sections.

To test this hypothesis, we compare a standard embedding-based retriever with our proposed graph-guided model using the same data and resources. The baseline retriever is a standard dense-vector search using cosine similarity over Chroma embeddings. During query time, the baseline retrieves the top-k results solely based on cosine similarity, without considering any metadata or structural information. This approach reflects current practices in many RAG systems and provides a strong reference point.

We associate each page with its corresponding chapter and section labels and create a directed metadata graph that encodes parent–child and sibling relationships. At query time, the system first retrieves top-k candidates using the baseline vector search. These candidates are mapped to their corresponding metadata nodes, expanded through neighbors in the graph (depth limited), and re-ranked using a weighted combination of embedding similarity and structural proximity.

5 Experiment and result

5.1 Experimental Setup

All experiments use the Psychology 2e corpus and environment described in Section 3. The textbook is segmented page-by-page and further split into overlapping chunks, which are embedded and indexed once. At query time we evaluate two retrieval systems:

- **Baseline embedding retriever**: pure vector search over chunks, without using any metadata or graph structure.

- **Graph-guided retriever**: the same vector search followed by metadata-graph expansion and re-ranking as described in Section 4.

We consider two tasks:

Method	Recall@5	Hit@5	MRR	Gold
Baseline (vector)	0.14	0.53	0.53	0.14
Graph-guided	0.34	0.66	0.57	0.34

Table 1: Chunk-level retrieval performance on all 32 QA queries.

- **Fine-grained question answering**: 32 human-labeled questions with chunk-level relevance annotations. Retrieval operates at the chunk level with $k = 5$.

- **Chapter-level summarization**: 12 high-level prompts annotated with relevant pages. Retrieval operates at the page level with $k = 20$.

For both tasks, we compute recall@k, hit rate, MRR, and gold-ratio@k using the definitions in Section 3.2. For the ablation study, we vary the graph expansion depth $d \in \{0, 1, 2, 3\}$ while keeping all other components fixed.

5.2 Retrieval Quality on Question Answering

We evaluate both retrievers on the full set of 32 fine-grained QA questions annotated with chunk-level relevance labels. Each system returns the top–5 chunks for each query. Table 1 reports recall@5, hit@5 (the fraction of queries with at least one correct chunk in the top–5), MRR, and the gold-ratio@5 (fraction of all gold chunks successfully retrieved).

Across all 32 QA queries, the graph-guided retriever substantially improves coverage of relevant content compared to pure vector search. Recall@5 more than doubles (0.14 → 0.34), and gold-ratio@5 shows an identical gain, indicating that the graph-based expansion consistently introduces additional relevant chunks that the vector retriever misses. Hit@5 also improves from 0.53 to 0.66, meaning a larger fraction of queries receive at least one correct supporting chunk in the top–5. MRR increases moderately (0.53 → 0.57), suggesting that relevant chunks not only appear more often but tend to rank slightly higher. Overall, these results show that the metadata-aware graph structure provides a meaningful advantage over standard embedding-based retrieval for fine-grained textbook question answering.

5.3 Retrieval Quality on Chapter-Level Summarization

We next evaluate the systems on 12 chapter-level summarization prompts, each annotated with the

Method	Recall@20	Hit@20	MRR	Gold
Baseline (vector)	0.14	0.40	0.22	0.14
Graph-guided	0.54	0.80	0.40	0.54

Table 2: Page-level retrieval performance on 12 summarization prompts.

Depth	R@5	MRR
0	0.14	0.53
1	0.47	0.54
2	0.34	0.57
3	0.21	0.54

Table 3: Ablation over graph expansion depth d on QA retrieval.

set of pages relevant to the concept being summarized. Retrieval operates at the page level with $k = 20$. As shown in Table 2, the metadata graph provides a substantial improvement over baseline vector search.

The summarization task shows even larger gains than fine-grained QA. The graph-guided retriever achieves a recall@20 of 0.54 compared to only 0.14 for the baseline, recovering nearly four times as many relevant pages. Hit@20 doubles (0.40 to 0.80), indicating that the graph-based method successfully places at least one correct page in the top–20 for most queries. MRR improves from 0.22 to 0.40, suggesting that relevant pages also appear earlier in the retrieved list. These results highlight the advantage of leveraging textbook structure for high-level conceptual queries, where chapter and section relationships encode strong contextual cues that pure embeddings fail to capture.

5.4 Ablation: Graph Expansion Depth

To assess the contribution of neighborhood expansion in the metadata graph, we vary the expansion depth $d \in \{0, 1, 2, 3\}$ on the QA task while holding all other retrieval components fixed. Depth 0 corresponds to graph re-ranking without exploring any neighboring nodes, while larger depths incorporate progressively broader sections of the chapter/section hierarchy.

As shown in Table 3, depth 0 performs poorly, showing that graph re-ranking alone is insufficient without incorporating structural neighbors. Depth 1 yields the highest recall (0.47), indicating that a small amount of structural expansion provides strong contextual cues. Depth 2 offers the best MRR (0.57), meaning relevant chunks tend to appear earlier in the ranking when expansion is moderately deep. At depth 3, performance degrades,

Method	Model	R@5	MRR	Hit@5
Baseline	MiniLM	0.14	0.53	0.53
Graph	MiniLM	0.34	0.57	0.66
Baseline	BGE	0.45	0.70	0.90
Graph	BGE	0.29	0.65	0.70

Table 4: Effect of embedding model choice on QA retrieval performance.

likely due to semantic drift into loosely related sections. Overall, shallow expansion (one or two hops) is most effective, confirming that local textbook structure is informative while overly broad expansion becomes noisy.

5.5 Latency

We measure end-to-end retrieval time over all 32 QA queries to quantify the computational overhead introduced by graph-guided retrieval. The baseline vector retriever requires 9.45 ms per query on average. The graph-guided retriever with expansion depth $d = 2$ requires 10.45 ms per query, corresponding to an overhead of approximately 1 ms (a 10.6% increase).

This latency increase is extremely small relative to the improvements in retrieval quality reported earlier. Both retrieval pipelines remain well under 15 ms per query, confirming that the proposed graph-guided approach is compatible with interactive applications where sub-second responsiveness is required.

Overall, the results demonstrate that incorporating textbook structure through a lightweight metadata graph leads to consistent and substantial improvements in retrieval quality across both fine-grained and high-level tasks. The graph-guided retriever recovers more relevant content, ranks it more highly, and generalizes better to broad conceptual prompts than standard embedding search. Ablation studies confirm that shallow neighborhood expansion is the most effective regime, while latency measurements show that these gains come at negligible computational cost. Together, these findings suggest that structural signals encoded in educational materials can meaningfully enhance retrieval components within RAG systems without sacrificing efficiency.

6 Discussion

The experimental results provide strong evidence that incorporating textbook structure through a metadata graph can improve retrieval quality in retrieval-augmented generation systems. Across

Method	R@5	MRR	Hit@5	Gold
Baseline (vector)	0.06	0.20	0.20	0.20
Graph-guided	0.32	0.36	0.60	0.70

Table 5: Page-level retrieval performance on the summarization task with a small retrieval budget ($k = 5$).

both QA and summarization tasks, the proposed graph-guided retriever consistently retrieved more relevant content and demonstrated higher contextual grounding than pure embedding-based retrieval. These outcomes support our original hypothesis: hierarchical structure in educational materials carries meaningful relational information that dense embeddings alone do not fully capture.

6.1 Effect of embedding model choice

Switching the embedding model from `all-MiniLM-L6-v2` to `bge-small-en-v1.5` produced a notable shift in performance. As shown in Table 4, the baseline retriever improved dramatically under BGE, achieving substantially higher recall and MRR than under MiniLM. Surprisingly, this improvement was large enough that the baseline surpassed the graph-guided retriever when using BGE. This reversal highlights a key limitation of our current design: the graph edge weights were manually selected based on intuition and tuned implicitly for the similarity geometry of MiniLM. Because BGE produces a different similarity distribution—more strongly aligned with semantic relevance—the same heuristic weights no longer reflect the correct balance of structural and semantic cues. As a result, graph expansion under BGE can introduce noise or surface overly broad neighbors, causing the graph-guided retriever to lose its advantage. This sensitivity suggests that fixed, human-designed graph weights are not robust across embedding models and should ideally be learned or adapted automatically.

6.2 Effect of k in the summarization task

Summary-style queries often require retrieving multiple pages because relevant information spans entire sections or chapters. Our experiments confirm that retrieval quality is sensitive to the choice of k (Table 5). With a small budget of $k = 5$, performance is low for both methods, but the graph-guided retriever still provides a substantial advantage: recall increases from 0.06 to 0.32, hit rate from 0.20 to 0.60, and gold coverage from 0.20 to 0.70. As k increases, both systems improve, but the graph-guided method benefits disproportionately. At $k = 20$, recall rises from 0.14 to 0.54, and hit rate doubles from 0.40 to 0.80, indicating that the metadata graph becomes more effective when the retrieval budget is large enough to include structurally adjacent pages. Together, these results show that summary tasks require a sufficiently large k to capture the distributed nature of high-level content, and that graph expansion amplifies these gains by propagating structural cues across related sections and chapters.

6.3 Unexpected observations

Two findings were especially unexpected. First, expansion depth behaved differently across metrics: depth 1 achieved the highest recall, while depth 2 produced the best MRR. This suggests that shallower expansion provides strong coverage, whereas moderate expansion improves ranking by locating more focused context. Second, the graph-guided retriever added almost no latency (approximately 1 ms per query), making the approach far more lightweight than anticipated.

6.4 Limitations

The method relies on clean, well-structured metadata extracted from the PDF outline. Documents without reliable chapter or section information, or with inconsistent formatting, may result in poorly constructed graphs. More importantly, the use of hand-crafted graph weights introduces subjectivity and reduces generalizability. As shown with the BGE experiments, these manually chosen weights interact strongly with the embedding distribution, and therefore may not transfer across corpora, domains, or embedding models.

6.5 Future directions

Several extensions follow naturally from these limitations. A promising direction is to learn graph edge weights automatically, either through supervised tuning using relevance labels, through contrastive learning, or through reinforcement learning with retrieval feedback. Another direction is to make graph expansion adaptive—adjusting depth based on local graph structure or query uncertainty. Finally, applying this approach to larger and more diverse corpora would test scalability and robustness, and integrating graph-aware retrieval with LLM-based re-ranking may yield additional improvements.

7 Conclusion

This project demonstrates that incorporating textbook structure through a lightweight metadata graph can meaningfully improve retrieval quality in RAG systems. By leveraging chapter and section relationships, the graph-guided retriever provides more complete and better-ranked evidence for both fine-grained questions and high-level summaries, while adding almost no computational overhead. However, experiments with alternative embedding models reveal that manually chosen graph weights are not robust and may fail when the embedding geometry changes. These findings suggest that structural metadata is a valuable signal for educational retrieval tasks, but future work should focus on learning or adapting graph weights automatically to ensure reliability across different models and corpora. The implementation and code for this work are publicly available at https://github.com/shiyuan-qiao/final_proj.git

8 Team Contribution

Shiyuan Qiao: Designed the model architecture, prepared the data, and defined the evaluation metrics Hit and Gold. Conducted experiments, summarized the results, and wrote the Experiment and results, Discussion, and Conclusion sections of the report.

Cloris Liu: Set up the LLM environment, designed the logic graph, implemented the retrieval module, and defined the evaluation metrics Recall and MRR. Wrote the Abstract, Introduction, Data, and Methodology sections of the report.

Jeremy Piperni: Prepared the query dataset. Created the presentation slides. Structured the presentation content and recorded the video demonstration.

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